

1 / 5 | Ages 9-12 | 15 minutes

Scale a colour recipe

What we are practising

Scale both parts of a ratio by the same factor.

What you need

Paper / Two coloured pencils or labelled symbols

Use B and Y labels instead of colours.

Before you begin

Write a pretend recipe: 2 blue scoops and 3 yellow scoops make one batch.

This uses drawn scoops only; no paint mixing or tasting is needed.

Try it together

1. Draw the five scoops in one batch.
2. Find how many complete batches make 20 scoops.
3. Calculate the blue and yellow amounts for that total.
4. Check that the parts sum to 20 and keep the original ratio.

Talk, draw or show your idea

Would adding the same number to both parts preserve the recipe?

Space to explore

2 / 5 | Ages 9-12 | 20 minutes

Chance versus results

What we are practising

Distinguish theoretical probability from observed frequency.

What you need

Six equal paper cards / Opaque bag or bowl with cloth cover / Pencil

An adult shuffles the cards face down and offers a blind selection.

Before you begin

Label three cards A, two B and one C; fold them identically.

Use large paper cards and keep bags away from faces and younger children.

Try it together

1. Predict the chance of drawing each letter.
2. Mix, draw one card without looking, record it and put it back.
3. Repeat for 18 draws, remixing every time.
4. Compare counts with the predicted proportions.

Talk, draw or show your idea

Must A appear exactly nine times in eighteen draws?

Space to explore

3 / 5 | Ages 9-12 | 15 minutes

Which notebook offer?

What we are practising

Calculate unit prices and distinguish cheapest per item from affordable today.

What you need

Paper / Pencil

Draw groups of notebooks and distribute pretend price tokens.

Before you begin

Write two offers: 3 notebooks for 12 tokens; 5 notebooks for 18 tokens. Budget: 15 tokens.

Use fictional tokens; this activity requires no purchase or payment details.

Try it together

1. Calculate the price per notebook for each bundle.
2. Identify which bundle has the lower unit price.
3. Decide which complete bundle fits the budget; bundles cannot be split.
4. Explain why the two answers differ.

Talk, draw or show your idea

Can the better unit price still be unaffordable?

Space to explore

4 / 5 | Ages 9-12 | 20 minutes

Same area, different fence

What we are practising

Distinguish area from perimeter using factor pairs.

What you need

Squared paper or a drawn grid / Pencil

Draw labelled rectangles without exact grid squares.

Before you begin

Imagine a garden of 24 square units with whole-number side lengths.

Use paper plans only; no fencing, digging or outdoor construction is involved.

Try it together

1. List the factor pairs of 24 to find possible rectangles.
2. Draw or label each rectangle and confirm its area.
3. Calculate each perimeter by adding all four sides.
4. Choose the rectangle needing the least fence among these options.

Talk, draw or show your idea

Why can equal areas have unequal perimeters?

Space to explore

5 / 5 | Ages 9-12 | 15 minutes

Which average tells the story?

What we are practising

Calculate two summaries and explain the effect of an unusually large value.

What you need

Paper / Pencil

Use drawn stacks of squares to represent the values.

Before you begin

Write five fictional reading-session lengths in minutes: 2, 3, 3, 4, 13.

Use fictional numbers; do not compare children's ability or worth using these measures.

Try it together

1. Add the five values and divide by five to find the mean.
2. Find the middle value in the ordered list to get the median.
3. Replace 13 with 3 and calculate both again.
4. Explain which summary changed more and why.

Talk, draw or show your idea

Does an average have to equal one of the actual measurements?

Space to explore

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

1. Scale a colour recipe

A little hint

Twenty scoops is four batches of five.

Answer / what to notice

Use 8 blue scoops and 12 yellow scoops; both original amounts are multiplied by 4.

Equal multiplication preserves the ratio; adding the same amount generally does not.

Make it easier

Scale the recipe to 10 scoops.

Take it further

Find the amounts for 7.5 scoops using fractional batches.

Another day, away from the screen

Invent a bead-pattern ratio on paper and scale it to two different totals.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

2. Chance versus results

A little hint

Each draw starts with all six cards again.

Answer / what to notice

A has chance $\frac{3}{6}$, B $\frac{2}{6}$ and C $\frac{1}{6}$. Counts near 9, 6 and 3 are plausible, but exact counts are not guaranteed.

Replacing and mixing restores the same chances; random short runs can differ from expected proportions.

Make it easier

Compare just two A cards and two B cards over eight draws.

Take it further

Combine results from several rounds and compare relative frequencies.

Another day, away from the screen

Design a different card mix and predict which outcomes become more likely.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

3. Which notebook offer?

A little hint

Divide the total price by the number of notebooks, then compare the full price with 15.

Answer / what to notice

The unit prices are 4 and 3.6 tokens. The five-pack is cheaper per notebook, but only the three-pack fits 15 tokens.

Unit price compares value for equal quantities; the total payment determines whether the budget covers the purchase.

Make it easier

Use 2 notebooks for 8 and 4 for 12.

Take it further

Find the lowest total cost to obtain at least eight notebooks using these whole bundles.

Another day, away from the screen

Invent two offers where the larger bundle has a worse unit price.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

4. Same area, different fence

A little hint

The distinct pairs are 1×24 , 2×12 , 3×8 and 4×6 .

Answer / what to notice

Their perimeters are 50, 28, 22 and 20 units; the 4 by 6 rectangle needs the least fence.

Area counts the inside squares; perimeter measures the outside boundary, which changes with shape.

Make it easier

Compare only 3 by 8 and 4 by 6.

Take it further

Repeat for area 36 and compare the most compact whole-number rectangle.

Another day, away from the screen

Look for two arrangements of twelve paper squares with different perimeters.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

5. Which average tells the story?

A little hint

The first total is 25; the middle of five ordered values is the third.

Answer / what to notice

Original mean 5, median 3. Revised values 2, 3, 3, 3, 4 have mean 3 and median 3.

The mean uses every value's size; the median depends on the ordered middle position, so one large value affects them differently.

Make it easier

Find only the median using five number cards.

Take it further

Create a five-number dataset with median 4 and mean 6.

Another day, away from the screen

Compare mean and median for invented game scores and explain what each reveals.